BM20BC module

File: LeoPatch_MCU.kicad_sch

Power

File: LeoPatch_Power.kicad_sch

USB-C

File: LeoPatch_USB.kicad_sch

ECG front end

File: LeoPatch_ECG.kicad_sch

LEDs + button

File: LeoPatch_UI.kicad_sch

ECG AD8232

File: LeoPatch_ECG_AD8232.kicad_sch

4xAA -> on/off -> reverse-polarity diode -> MCP1703A 3.3 V LDO -> BM20BC + AD8232 (USB-C 5 V also feeds the LDO)

ECG electrodes -> 3.5 mm jack -> AD8232 -> ECG_OUT -> nRF54LM20B AIN4 (P1.06)

Result LEDs, EXPORT button, SWD header (first flash), USB-C (updates + data). 4-layer, standard vias.

Rice ELEC 494 — LeoPatch

Sheet: /
File: LeoPatch_v4.kicad_sch

Title: LeoPatch v4 — newborn CCHD ECG screening (BM20BC, MAX30003 + A

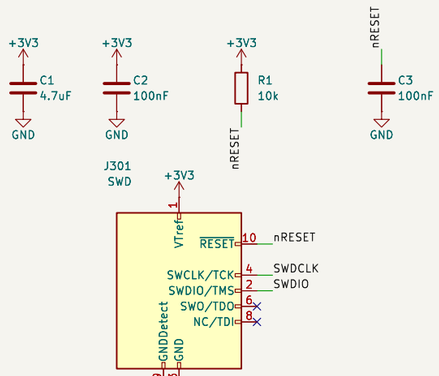
Size: A4	Date: 2026-11-12	**Rev: 4.0**
KiCad E.D.A. 9.0.7		Id: 1/7

MCU: Fanstel BM20BC module — nRF54LM20B (Arm M33 + Axon NPU), 32 MHz + 32 kHz crystals, DC/DC, chip antenna

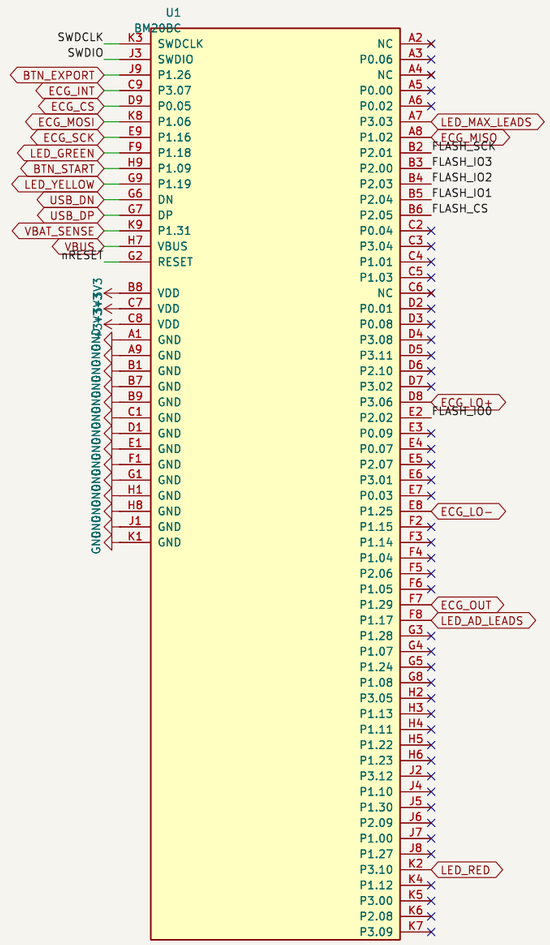
Antenna end of the module sits at the board edge over a copper keep-out (Fanstel mounting guideline).

USB: VBUS (H7), D- (G6), D+ (G7) go to the USB-C sheet. SWDIO, RESET and USB use vias between module pads.

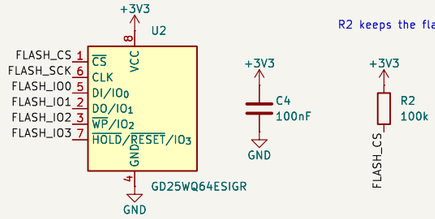
Decoupling, reset, SWD (as Fanstel EV-BM20)



First flash only (MCUboot bootloader) with the nRF54LM20 DK "Debug out" 10-pin cable; after that, update over USB-C or BLE.

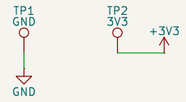


External flash: GD25WQ64E (8 MB) on P2.00–P2.05, same pins as the nRF54LM20 DK flash.
Place C4 next to U2 pin 8.



R2 keeps the flash deselected while the MCU pins float (reset, programming).

Test points (1 mm pads on the top side): GND and 3V3



SW303 pulls nRESET low. R1 (10k) and C3 (100nF) are the pull-up and debounce.



Rice ELEC 494 — LeoPatch

Sheet: /BM20BC module/
File: LeoPatch_MCU.kicad_sch

Title: LeoPatch v2 — Fanstel BM20BC (nRF54LM20B + NPU)

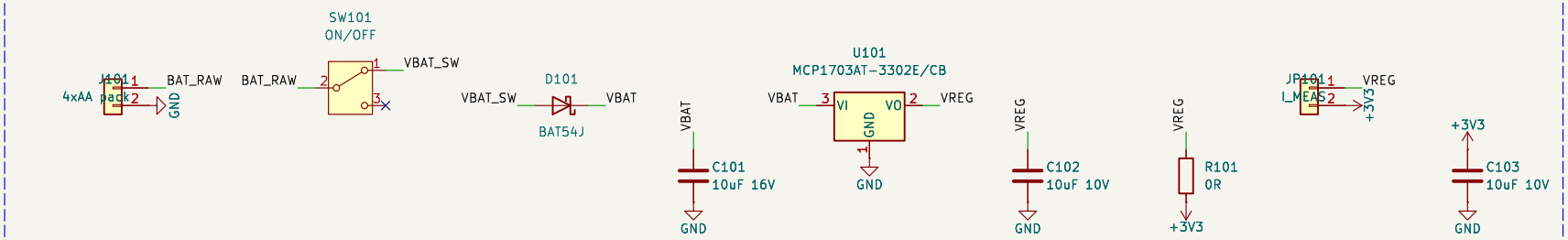
Size: A3	Date: 2026-10-25	**Rev: 2.0**
KiCad E.D.A. 9.0.7		Id: 2/7

POWER: 4xAA pack -> on/off switch -> reverse-polarity diode -> 3.3 V LDO

4xAA = 3.6 V (depleted) .. 6.4 V (fresh alkaline). MCP1703A: 16 V max in, 2 uA Iq, 250 mA.

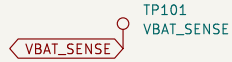
USB-C 5 V joins VBAT through D102 (diode-OR with the battery path); D101 blocks it from the cells.

Battery input + regulator



Current measurement: remove R101, connect PPK2 across JP101.

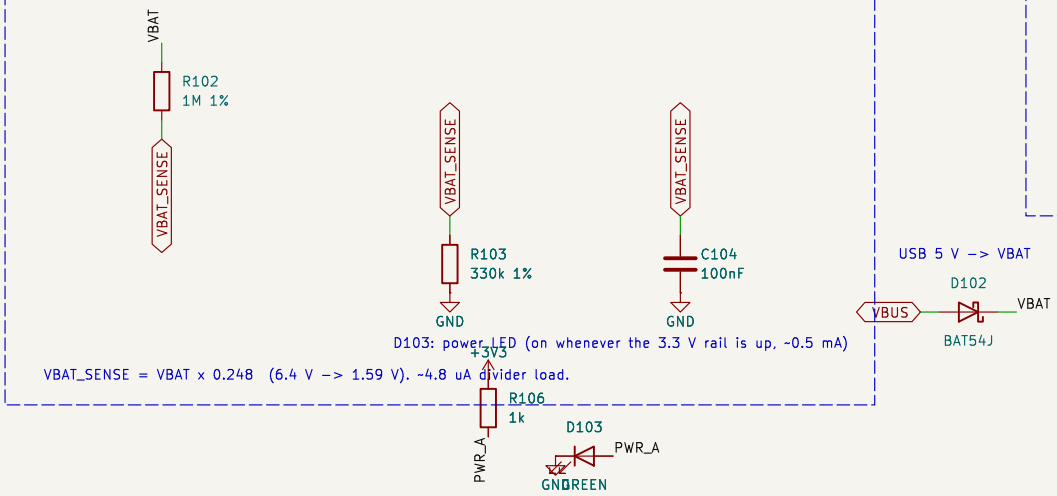
Test points (bare 1.5 mm pads, top side,



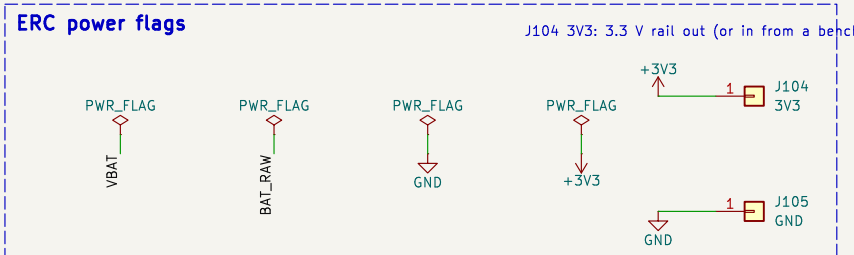
J103 VIN: external supply in, in parallel with the batteries and USB (
so it cannot back-feed either). 3.6 – 6 V. Clip the supply's minus to



Battery voltage monitor -> BM20BC K9 P1.31 (AIN1)



ERC power flags



J104 3V3: 3.3 V rail out (or in from a bench supply: pull the JP101

Rice ELEC 494 – LeoPatch

Sheet: /Power/

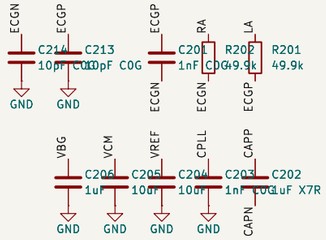
File: LeoPatch_Power.kicad_sch

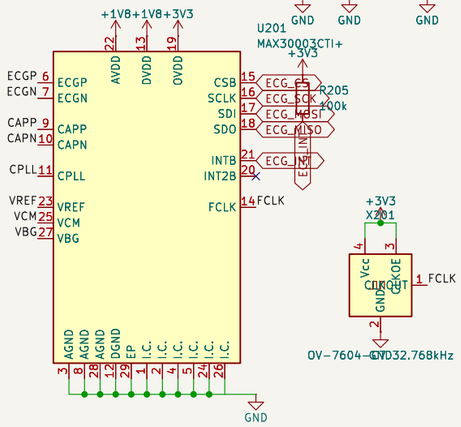
Title: LeoPatch – Power (4xAA -> 3.3 V)

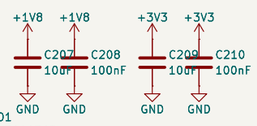
Size: A4	Date: 2026-10-25	Rev: 2.0
KiCad E.D.A. 9.0.7		Id: 3/7

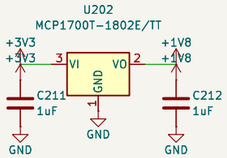
ECG FRONT END: MAX30003, two-electrode (Figure 11 of the datasheet).
ECGP = LA, ECGN = RA (lead I). The third electrode (RL) is not used by the chip; its jack and header pins are only tied together.

Input network as on the MAX30003WING2 evaluation kit: 49.9k pulse-proof series resistors, C201 1 nF across ECGP-ECGN, C213 / C214 10 pF to GND (matched C0G), R208 499k from VCM to the RL electrode. C202 1 uF = 0.5 Hz high-pass.

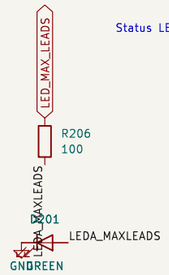




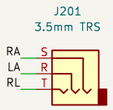




Status LED D201 LEADS: firmware lights it while both electrodes are on (DCLOFFINT clear).









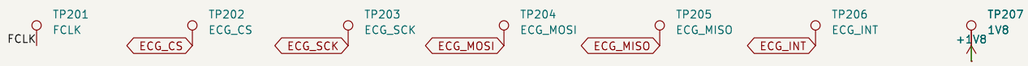
Lead cable: 3.5 mm TRS plug, Tip = RL, Ring = LA, Sleeve = RA. J202 = backup solder pads.

C202 sets the analog high-pass: 10 uF = 0.04 Hz (waveform quality), 1 uF = 0.4 Hz (more motion rejection).
Gain (20/40/80/160), 0.5 Hz digital high-pass and 40/100/150 Hz low-pass are set over SPI.

AVDD/DVDD 1.8 V from U202; OVDD = 3.3 V so the SPI lines match the module. FCLK must be 32.768 kHz (X201).

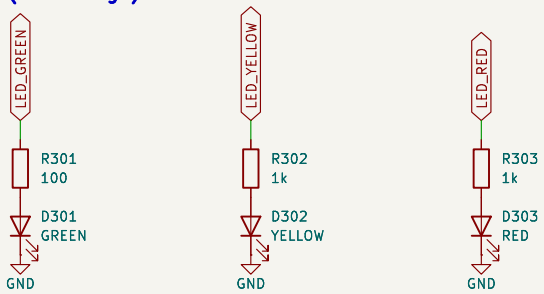
Module pins: SCK P1.16 (E9), MOSI P1.06 (K8), MISO P1.02 (A8), CS P0.05 (D9), INT P3.07 (C9).

Test points (bare 1.5 mm pads, top side, not in the BOM)



Rice ELEC 494 – LeoPatch		
Sheet: /ECG front end/		
File: LeoPatch_ECG.kicad_sch		
Title: LeoPatch – ECG front end (MAX30003)		
Size: A3	Date: 2026-11-11	**Rev: 3.0**
KiCad E.D.A. 9.0.7		Id: 4/7

Result LEDs (active high)

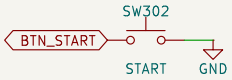


Green = signal OK / normal, Yellow = needs review, Red = abnormal / bad signal

Export button



Export the recording over BLE (P1.26, internal pull-up)



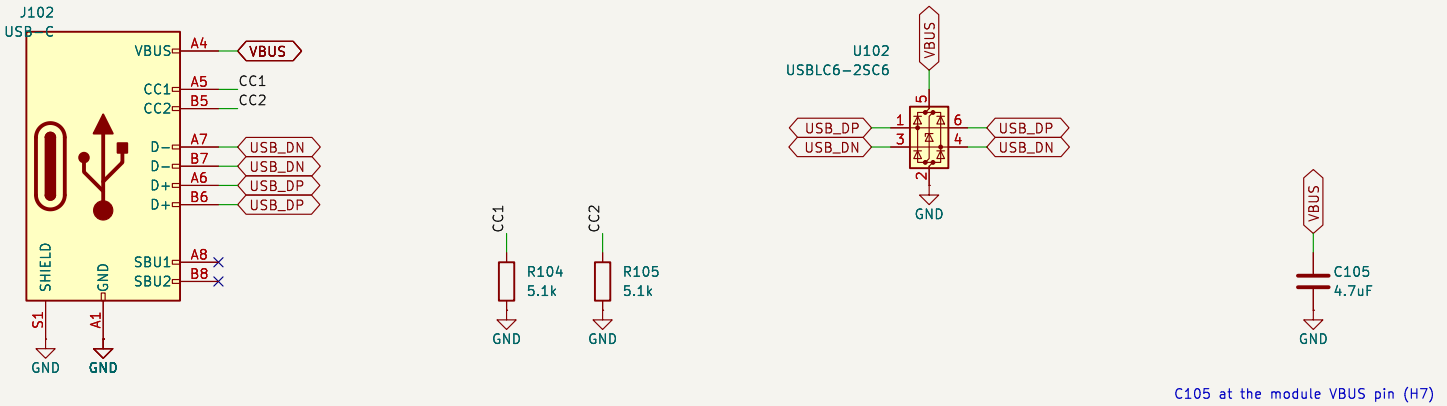
Rice ELEC 494 – LeoPatch		
Sheet: /LEDs + button/ File: LeoPatch_UI.kicad_sch		
Title: LeoPatch v2 – Result LEDs and export button		
Size: A4	Date: 2026-10-25	**Rev: 2.0**
KiCad E.D.A. 9.0.7		Id: 5/7

USB-C: USB 2.0 device to the nRF54LM20B USB controller, plus 5 V into the regulator

5.1k on CC1/CC2 = sink (device). USBLC6-2SC6 ESD on D+/D-/VBUS, right at the connector.

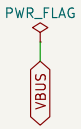
SAFETY: do not plug into a mains-powered computer while electrodes are on a patient (IEC 60601).

Connector, CC pull-downs, ESD, VBUS



C105 at the module VBUS pin (H7)

ERC power flag



Rice ELEC 494 – LeoPatch		
Sheet: /USB-C/		
File: LeoPatch_USB.kicad_sch		
Title: LeoPatch v2 – USB-C (update, data export, 5 V power)		
Size: A4	Date: 2026-10-25	**Rev: 2.0**
KiCad E.D.A. 9.0.7		Id: 6/7

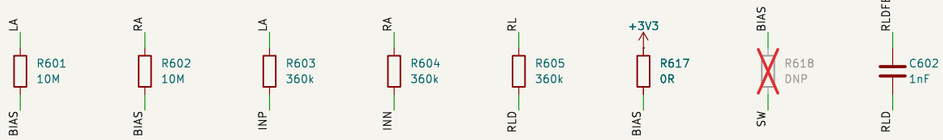
ECG FRONT END: AD8232 — component values copied 1:1 from SparkFun AD8232 Heart Rate Monitor (CC BY-SA 4.0)

Same circuit as the red AD8232 breakout modules the team is testing; their on-board heartbeat LED (D1/R16/SJ4) is omitted.

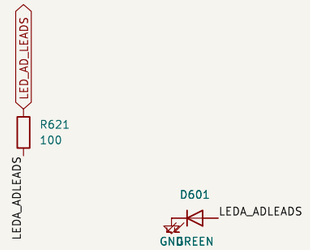
Lead cable: 3.5 mm TRS plug, Tip = RL (yellow), Ring = LA (green), Sleeve = RA (red) — same as module.

Module pads: ECG_OUT -> F7 (P1.29 / AIN3), LO+ -> D8 (P3.06), LO- -> E8 (P1.25), SDN -> A7 (P3.03). TP601 = ECG_OUT test pad.

Input bias / protection / right-leg drive

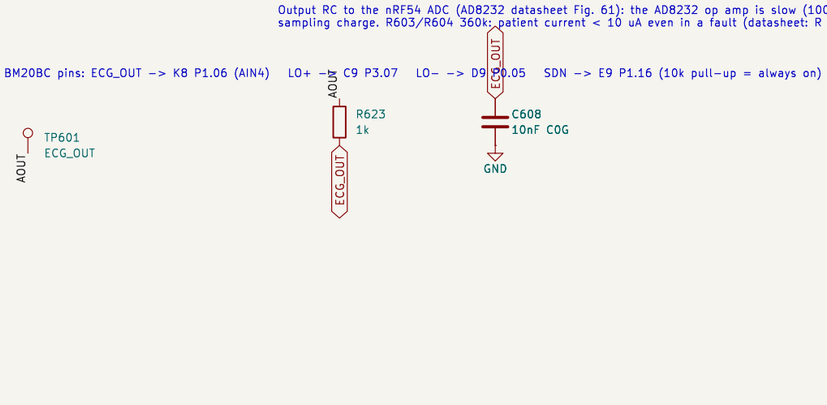


Status LED D601 LEADS: firmware lights it while LO+ and LO- are both low (both electrodes on).

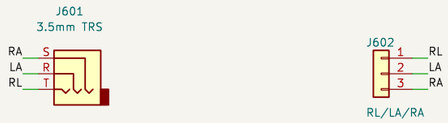


Output RC to the nRF54 ADC (AD8232 datasheet Fig. 61): the AD8232 op amp is slow (100 kHz GBP), C608 supplies the ADC sampling charge. R603/R604 360k: patient current < 10 uA even in a fault (datasheet: R >= supply / 10 uA).

BM20BC pins: ECG_OUT -> K8 P1.06 (AIN4) LO+ -> C9 P3.07 LO- -> D9 P0.05 SDN -> E9 P1.16 (10k pull-up = always on)



Electrode connectors

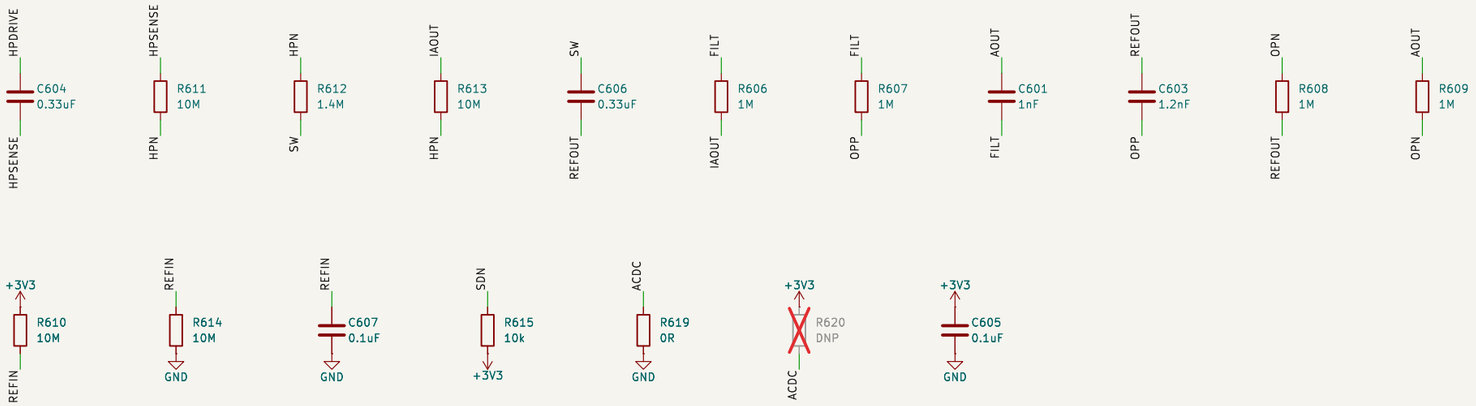


J602 = backup solder pads (same as module RA/LA/RL holes)

Test points (bare 1.5 mm pads, top side, not in the BOM)



High-pass (0.5 Hz) / low-pass (~145 Hz) filter network, total gain ~200, reference, control



R219 fitted = AC lead-off mode (as SparkFun). R217 fitted / R218 DNP = inputs biased as SparkFun.

Rice ELEC 494 — LeoPatch

Sheet: /ECG AD8232/

File: LeoPatch_ECG_AD8232.kicad_sch

Title: LeoPatch v4 — ECG front end B (AD8232)

Size: A3	Date: 2026-11-12	**Rev: 4.0**
KiCad E.D.A. 9.0.7		Id: 7/7